

The quadratic formula



1 Considering the quadratic formula, find the values of a , b and c in the following quadratic equations:

a $x^2 + 7x + 10 = 0$

b $x^2 - 3x - 4 = 0$

c $2x^2 + 9x = 0$

d $4x^2 + 3x = 5$

e $8x^2 - 3x = 0$

f $-8x^2 + 3x = 0$

g $-2x^2 + 9x + 5 = 0$

h $3x^2 - 8x + 2 = 9x - 7$

2 Solve the following equations using the quadratic formula:

a $x^2 + 5x + 6 = 0$

b $x^2 - 5x + 6 = 0$

c $x^2 - 5x + \frac{9}{4} = 0$

d $x^2 - 8x + \frac{55}{4} = 0$

e $2x^2 + 6x - 8 = 0$

f $4x^2 - 10x + 4 = 0$

g $4x^2 + 7x + 3 = 0$

h $4x^2 - 17x - 15 = 0$

i $-6 + 7x + 5x^2 = 0$

j $-6 - 13x + 5x^2 = 0$

k $-20 - 11x + 3x^2 = 0$

l $-20 + 21x + 5x^2 = 0$

3 Solve the following equations, leaving your answer in surd form:

a $x^2 - 7x + 9 = 0$

b $x^2 - 5x - 2 = 0$

c $-2x^2 - 15x - 4 = 0$

d $3x^2 + 9x - 4 = 0$

e $5x^2 - 15x + 2 = 0$

f $-5x^2 - 15x + 3 = 0$

g $5x = (x - 5)(3x + 3)$

h $\frac{3x + 1}{3x - 1} - \frac{3x - 1}{3x + 1} = 5$

4 Consider the equation $2x^2 = 14$.

a Solve for x using the quadratic equation. Give your answer in surd form.

b Give x as decimal number correct to three decimal places.

5 Solve the following equations, rounding your answers to three decimal places:

a $x^2 + 3x - 6 = 0$

b $x^2 + 7x - 3 = 0$

c $4x^2 + 7x + 2 = 0$

d $5x^2 + 9x + 2 = 0$

e $2x(x - 4) = 3x + 1$

f $3x(x + 4) = -3x + 4$

g $7.1x^2 + 5.3x - 1.5 = 0$

h $1.8x^2 + 5.2x - 2.3 = 0$

- 6 Solve the following expression for m :



$$10 - 6m + 2m^2 = m^2 + 8m + 9$$

- 7 For each of the given solutions to a quadratic equation:

- i Find the values of a , b and c .
- ii Write down the quadratic equation that has these solutions.

a
$$x = \frac{-5 \pm \sqrt{5^2 - 4 \times (-7) \times 10}}{2 \times (-7)}$$

b
$$x = \frac{-22 \pm \sqrt{484 - 108}}{18}$$

Applications

- 8 An object is launched from a height of 80 ft with an initial velocity of 107 ft/s.

After x seconds, its height, h , is given by:

$$h = -16x^2 + 107x + 80$$

Find the number of seconds, x , after which the object is 30 ft above the ground. Round your answer to one decimal place.

- 9 An object is launched from a height of 8 m with an initial velocity of $k \text{ ms}^{-1}$.

After t seconds, its height, $h(t)$, is given by:

$$h(t) = -2t^2 + kt + 8$$

The object was in flight for 4 seconds before reaching the ground. What was its initial velocity k ?